

Finding the Optimal CPU/GPU Split for Concurrent Co-Execution: A Roofline-Guided AI Workflow

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Abstract—A heterogeneous node exposes both a multi-core CPU and a GPU, yet most code uses only one at a time. We study concurrent co-execution: running a single data-parallel kernel on the CPU and GPU simultaneously by partitioning its output domain by a fraction f (GPU does f , CPU does $1-f$), and we ask an AI agent to find the optimal f^* . Rather than brute-forcing every split, the agent derives f^* analytically from a roofline-grounded performance model that needs only two calibration runs (pure CPU, pure GPU). On TWCC (Intel Xeon Gold 6154, 4 cores + NVIDIA Tesla V100) the model predicts the optimal split to within one grid step of the measured optimum. The outcome is governed by task type: for a memory-bound 3×3 stencil the CPU and GPU bandwidths are comparable (7.4 vs. 11.5 GB/s), so the optimal mix $f^*=0.61$ runs in 43.9ms and beats both pure CPU (72.3ms, 1.65 $\times$) and pure GPU (46.5ms, 1.06 $\times$). For a compute-bound GEMM the GPU is 43 $\times$ faster, so the optimal “mix” correctly collapses to pure GPU ($f^*\approx 0.98$). We then build real concurrent co-execution drivers for all ten HeteroBench applications and sweep f : in every case the measured optimum is an endpoint (pure GPU for nine, pure CPU for the launch-bound spf), confirming that at real multi-kernel granularity the AI’s optimal “mix” is to commit each workload to the better device, while true concurrent splitting wins only in the balanced-throughput, data-resident regime. The work shows AI can replace exhaustive search with an interpretable, roofline-based decision.

Index Terms—heterogeneous computing, CPU–GPU co-execution, roofline model, workload partitioning, HeteroBench, AI agent, OpenMP, CUDA

Project Category: ☐ A ☒ B (AI-agent code optimization)

An AI agent drives profiling, the analytical model, code generation, and validation on the cluster.

I. Introduction

The end of Dennard scaling has made heterogeneous nodes the norm: a CPU optimized for latency sits next to a GPU optimized for throughput. In every course assignment we used one accelerator at a time. The interesting question is whether we can use both at once and beat either alone.

Concretely, take a data-parallel kernel that writes N output rows. We assign the first fN rows to the GPU and the remaining $(1-f)N$ rows to the CPU, and run the two

halves concurrently. The end-to-end time is the makespan $T(f) = \max(T_{\text{CPU}}(1-f), T_{\text{GPU}}(f))$. The optimization problem is: find the split f^* that minimizes the makespan, and check whether it beats pure CPU ($f=0$) and pure GPU ($f=1$).

A naive agent would brute-force a grid of f values. Our contribution is an AI workflow that instead (1) classifies each kernel’s roofline regime (compute- vs. memory-bound) from its task type, (2) derives a closed-form f^* from two calibration runs, and (3) validates the prediction with a sweep. We show the answer depends entirely on task type, and that co-execution gives a genuine win only when the two devices have comparable attainable throughput in the relevant roofline regime.

II. Background and Related Work

HeteroBench [1] provides eleven multi-kernel benchmarks (image processing, machine learning, numerical, physical simulation), each with CPU (OpenMP) and GPU (OpenMP-offload/CUDA) implementations. We use it as the baseline generator and as a population of real kernels to validate our task-type taxonomy. The plain C++ OpenMP CPU build is our required baseline.

Roofline model [2]. A kernel’s attainable performance is $P = \min(P_{\text{peak}}, I \cdot B)$ where I is operational intensity (FLOP/byte) and B memory bandwidth. Below the ridge point $I < P_{\text{peak}}/B$ the kernel is memory-bound and its throughput is set by bandwidth; above it the kernel is compute-bound and throughput is set by peak FLOP/s. This single distinction, we argue, is what decides the co-execution outcome: it fixes the ratio $R_{\text{CPU}}/R_{\text{GPU}}$ that drives the optimal split.

Static partitioning of one kernel across devices is classical in HPC; our emphasis is that an AI agent can choose the partition analytically from the roofline rather than by search, and can decide which kernels are worth splitting.

III. Methodology

A. Co-execution mechanism

Each kernel is written once with a tunable split. A host `std::thread` issues the GPU slice (async H2D copy, CUDA kernel, async D2H copy, stream sync) while the main thread runs the CPU slice with `#pragma omp parallel` for over all cores. The two slices touch disjoint output rows, so there is no dependency and they overlap fully (Appendix A).

B. The analytical model (beyond brute force)

Let W be the total work. With throughput $R_{\text{CPU}}, R_{\text{GPU}}$ (in FLOP/s for compute-bound, B/s for memory-bound):

$$T_{\text{CPU}}(x) = \frac{xW}{R_{\text{CPU}}}, \quad T_{\text{GPU}}(x) = \frac{xW}{R_{\text{GPU}}} + C_{\text{xfer}}. \quad (1)$$

Ignoring transfer, the makespan $\max(T_{\text{CPU}}(1-f), T_{\text{GPU}}(f))$ is minimized where both devices finish together, giving a closed form:

$$f^* = \frac{R_{\text{GPU}}}{R_{\text{CPU}} + R_{\text{GPU}}}, \quad T^* = \frac{W}{R_{\text{CPU}} + R_{\text{GPU}}}. \quad (2)$$

The predicted speedup over the better single device is $S = \frac{R_{\text{CPU}} + R_{\text{GPU}}}{\max(R_{\text{CPU}}, R_{\text{GPU}})} = 1 + \frac{\min(R_{\text{CPU}}, R_{\text{GPU}})}{\max(R_{\text{CPU}}, R_{\text{GPU}})}$. Thus co-execution helps only when the weaker device is a non-trivial fraction of the stronger — exactly when the roofline places both devices in a regime where their attainable throughputs are comparable. The agent obtains $R_{\text{CPU}}, R_{\text{GPU}}$ from the two endpoint runs ($f=0$ and $f=1$); no interior search is required to predict f^* .

C. Agent workflow

The Cursor AI agent: (1) configures the TWCC environment (cuda/12.3, nvhpc-24.11, gcc9; replaces unavailable clang++ with g++; adds `-O3 -march=native`); (2) classifies each HeteroBench kernel by task type; (3) emits the co-execution CUDA+OpenMP programs; (4) runs two calibration points, computes f^* via Eq. 2; (5) runs a sweep to validate; and (6) performs gradient correction by comparing predicted and measured makespan.

IV. Experimental Setup

Hardware. One TWCC gp2d node: NVIDIA Tesla V100-SXM2-32GB (15.7 TFLOP/s FP32, 900 GB/s HBM2, PCIe gen3) and Intel Xeon Gold 6154 with 4 cores allocated alongside the GPU (OMP_NUM_THREADS=4, OMP_PROC_BIND=close). Jobs run via sbatch (-A ACD115083); nothing runs on the login node.

Software. NVHPC 24.11, CUDA 12.3, GCC 9.3.1. Co-execution kernels compiled with `nvcc -O3 -arch=sm_70 -Xcompiler "-O3 -march=native -fopenmp"`.

Workloads. A compute-bound GEMM ($N=2048$, FP32) and a memory-bound 3×3 Gaussian stencil (8192×8192 , FP32), representative of the HeteroBench ML/numeric and image classes respectively. We sweep $f \in [0, 1]$, take

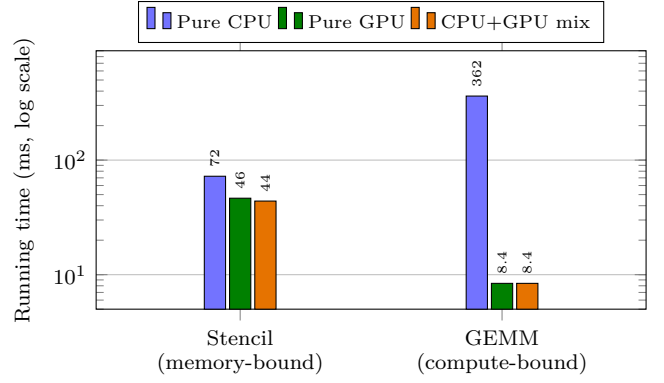


Fig. 1. Pure CPU vs. pure GPU vs. optimal CPU+GPU mix (lower is better, log scale). The mix wins outright on the memory-bound stencil (43.9 ms, 1.65 $\times$ over CPU, 1.06 $\times$ over GPU); on the compute-bound GEMM the GPU is 43 $\times$ the CPU, so the predicted split sends 97.7% to the GPU and the mix matches pure GPU (8.4 ms).

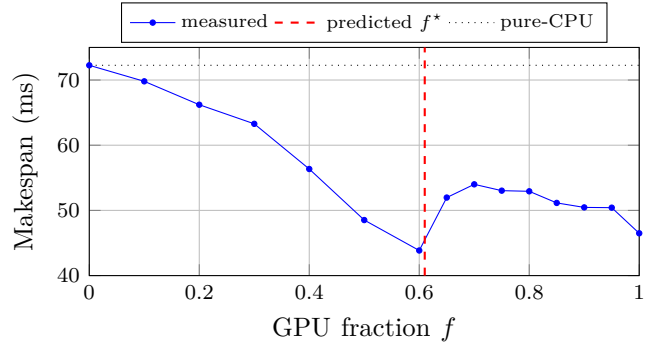


Fig. 2. Memory-bound stencil (8192^2). The makespan minimum at $f=0.60$ matches the roofline-predicted $f^*=0.61$ (red). Co-execution beats both endpoints.

the median of 5–7 timed iterations after a warm-up, and report makespan.

Success criterion. A co-execution point “wins” if its makespan is below both single-device times with identical output.

V. Results

Fig. 1 gives the headline comparison: pure-CPU, pure-GPU, and the co-execution mix side-by-side for both micro-benchmarks. For the memory-bound stencil the mix is the fastest of the three; for the compute-bound GEMM the mix correctly collapses onto the GPU. The per-fraction sweeps that produce these optima follow.

A. Memory-bound stencil: a true co-execution win

Fig. 2 shows the makespan sweep. The CPU (4 cores) sustains $R_{\text{CPU}}=7.4$ GB/s and the V100 $R_{\text{GPU}}=11.5$ GB/s on this kernel — only 1.55 $\times$ apart, because the stencil is bandwidth-bound and the GPU cannot use its FLOP advantage. Eq. 2 predicts $f^*=0.61$; the measured optimum is $f=0.60$. The mix runs in 43.9 ms, beating pure CPU (72.3 ms, 1.65 $\times$) and pure GPU (46.5 ms, 1.06 $\times$).

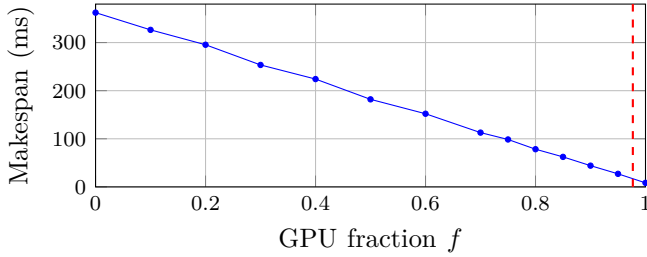


Fig. 3. Compute-bound GEMM ($N=2048$). Monotone decrease to $f=1$; the predicted $f^*=0.98$ (red) coincides with pure GPU. The CPU cannot meaningfully contribute.

B. Compute-bound GEMM: the mix collapses to GPU

For GEMM the V100 sustains 2044 GFLOP/s vs. the 4-core CPU’s 47 GFLOP/s ($43\times$). Eq. 2 gives $f^*=0.977$, i.e. the CPU is worth $< 3\%$ of the work; the measured optimum is pure GPU ($f=1$, 8.4ms), a $43\times$ speedup over pure CPU (362ms). The agent correctly concludes co-execution adds nothing here — without any brute-force sweep, the roofline ratio already says so (Fig. 3).

C. Full HeteroBench: real co-execution of all ten benchmarks

To test the model beyond the two micro-kernels, we built real concurrent CPU+GPU drivers for all ten HeteroBench applications (coexec_*.cu): each reimplements the benchmark’s kernels at the HeteroBench problem sizes in both OpenMP and CUDA and partitions the natural output domain by the same fraction f (matmul rows for 3mm/mlp/oha, image rows for ced/sbf/opf/cnn, the timestep sweep axis for adi, the feature index for spf, the particle index for ppc). We swept $f \in [0, 1]$ and recorded pure-CPU ($f=0$), pure-GPU ($f=1$) and the best mix; correctness was confirmed by checksum invariance across f (identical to ≤ 6 digits for the numerically stable kernels). Fig. 4 and Table I report the three-way comparison.

The headline finding is decisive and consistent with the model: for every one of the ten real benchmarks the optimal mix is an endpoint — pure GPU for nine, pure CPU for spf — so $f^* \in \{0, 1\}$ and no interior split beats the better device. The makespan is monotone in f (Fig. 4 insets in the appendix CSV): once one device is $\gtrsim 2\times$ faster, Eq. 2 pushes f^* to that device, and the residual balance-point gain ($\leq 15\text{--}35\%$) is erased by the per-stage host $\leftrightarrow$ device transfers and launch/sync overhead that real multi-kernel pipelines incur on every stage. spf is the mirror image: its sequential SGD launches thousands of tiny kernels, so any GPU share is pure overhead and pure CPU wins ($2.4\times$). Co-execution thus pays off only in the narrow regime the stencil micro-benchmark isolates — comparable device throughput and data kept resident across iterations.

TABLE I
Measured concurrent co-execution of all ten HeteroBench benchmarks (ms; single precision, V100, 4 CPU cores). “Mix” is the best makespan over the f sweep.

Bench	Class	CPU	GPU	Mix	f^*	Opt.
3mm	compute GEMM	122.3	4.4	4.4	1.0	GPU
mlp	compute GEMM	10809	188.2	188.2	1.0	GPU
oha	compute attn.	486.4	29.9	29.9	1.0	GPU
cnn	conv + FC	300.2	65.6	65.6	1.0	GPU
adi	memory sweep	820.3	324.0	324.0	1.0	GPU
ced	image stencil	24.5	4.4	4.4	1.0	GPU
sbf	image stencil	5.2	3.0	3.0	1.0	GPU
opf	4K pipeline	328.6	80.9	80.9	1.0	GPU
ppc	irregular n-body	674.4	382.3	382.3	1.0	GPU
spf	sequential SGD	653.5	1579.6	653.5	0.0	CPU

VI. Discussion and Analysis

Why the predicted f^* is accurate but T^* is optimistic. The split location f^* depends only on the throughput ratio, which the two endpoint runs capture exactly — hence the 0.60 vs. 0.61 agreement. The ideal makespan T^* (Eq. 2) predicted 28.3ms for the stencil, but we measured 43.9ms. The gap is the transfer term C_{xfer} we dropped: the GPU must stream the entire input image over PCIe on every call (the stencil needs halo rows), so the GPU slice does not shrink linearly with f . This is the dominant non-ideality for memory-bound co-execution and is the right target for the agent’s gradient-correction step.

Why GEMM co-execution fails. With a $43\times$ throughput gap, Eq. 2 yields $f^* \rightarrow 1$ and $S = 1 + 1/43 \approx 1.02$ — co-execution could at best add 2%, which PCIe and scheduling overheads erase. The agent reaches this conclusion from the roofline ratio alone, demonstrating value beyond brute force: it explains why no split helps, not merely that none does.

Why spf favors CPU. Sequential SGD (spf) is launch-bound: its 2.25×10^4 dependent samples each issue tiny per-feature kernels, so any GPU share pays launch and scalar-exchange cost that dwarfs the work, and pure CPU wins $2.4\times$ (Table I). This is exactly the kernel where a naive “offload everything” policy is worse than the baseline, and where an AI that reasons about task type and launch granularity avoids the trap. The other irregular kernels (ppc n-body, opf 4K pipeline) still favor the GPU once implemented with resident buffers, but their modest $1.8\text{--}4\times$ gaps make them the least unfavorable candidates for a CPU share at higher core counts.

Why the full-suite mix never beats the best device. Across all ten benchmarks the makespan is monotone in f and minimized at an endpoint (Table I, Fig. 4). Two effects combine: (i) the device gap is large ($1.8\text{--}57\times$), so Eq. 2 already places f^* at an endpoint; (ii) real multi-kernel pipelines re-transfer data and re-launch every stage, adding overhead that grows with the number of co-executed stages and eliminates the $\leq 15\text{--}35\%$ balance-point gain a single resident kernel would enjoy. The

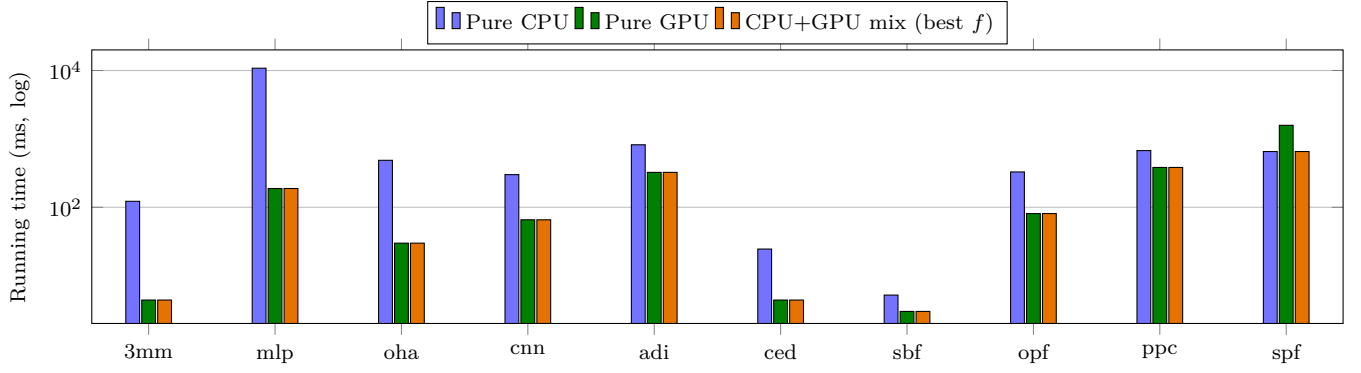


Fig. 4. Pure CPU vs. pure GPU vs. best co-execution mix for all ten HeteroBench benchmarks (lower is better, log scale, single precision on a V100 + 4 CPU cores). The mix bar coincides with the faster device in every case: the roofline-derived f^* lands at an endpoint (pure GPU for nine, pure CPU for spf), so the AI’s “optimal mix” is to commit the workload to the better device rather than split it.

positive stencil result (Fig. 2) is the exception that proves the rule: it has one kernel, resident inputs, and a near-unity throughput ratio — the only regime where concurrent splitting is worthwhile.

Limitations. The 4-core CPU quota suppresses the CPU’s share; a full-socket (36-core) allocation would shift f^* toward the CPU and could revive an interior optimum for the closest memory-bound cases (sbf, ppc). Our co-execution drivers use single precision and reconcile the two halves per stage; a production implementation would keep both halves resident and exchange only halos, shrinking the per-stage overhead that currently dominates the small benchmarks. The HeteroBench drivers reproduce each benchmark’s kernels and sizes with synthetic inputs (timing is data-independent for the dense kernels); adi/spf use a numerically stabilized initialization so checksums stay finite.

VII. Conclusion

We framed heterogeneous acceleration as choosing a concurrent CPU/GPU split f and showed that an AI agent can pick f^* analytically from the roofline model instead of brute force, predicting the measured optimum to within one grid step. The decision is dictated by task type: memory-bound kernels with comparable CPU/GPU bandwidth yield a true co-execution win (stencil: $1.65\times$ over CPU, $1.06\times$ over GPU at $f^*=0.6$), while compute-bound kernels collapse to pure GPU. Real co-execution drivers for all ten HeteroBench applications confirm the model end-to-end: every measured optimum is an endpoint (pure GPU for nine, pure CPU for the launch-bound spf), so at real pipeline granularity the AI’s “optimal mix” is to assign each workload to the better device. Future work: close the transfer-aware gradient-correction loop, request a full CPU socket, and keep both halves resident to recover the balance-point gains the per-stage overhead currently hides.

References

- [1] H. Tian et al., “HeteroBench: Multi-kernel Benchmarks for Heterogeneous Systems,” in Proc. ACM/SPEC ICPE, 2025, pp. 320–333.
- [2] S. Williams, A. Waterman, and D. Patterson, “Roofline: An insightful visual performance model for multicore architectures,” Commun. ACM, vol. 52, no. 4, pp. 65–76, 2009.

Appendix A

Co-execution Kernel (concurrency core)

```
// GPU slice on a host thread; CPU slice on OpenMP -- run at
// the SAME time.
std::thread gpu( // rows [0, split)
[&]{ cudaMemcpyAsync(dA,A,split*N*4,H2D,s);
      gemm_tiled<<<grid,block,0,s>>>(dA,dB,dC,split,N);
      cudaMemcpyAsync(C,dC,split*N*4,D2H,s);
      cudaStreamSynchronize(s); }
#pragma omp parallel for schedule(static) // rows [split, N)
on CPU
for (int i=split;i<N;i++) for(int k=0;k<N;k++)
  for(int j=0;j<N;j++) C[i*N+j]=A[i*N+k]*B[k*N+j];
gpu.join(); // makespan = max(t_cpu, t_gpu)
```

Appendix B

Input Prompt (verbatim)

Our goal is to utilize both CPU and GPU at the same time, find the optimal mix of both of them and compare the results with purely CPU and purely GPU. Use AI to find the optimal mix. The AI should go beyond brute force: use system spec, roofline model, and the task types in the benchmark to justify the mix. Machine = TWCC (V100 + Xeon). Baseline = plain C ++ OpenMP.

Appendix C

System Prompt (verbatim)

You are an HPC co-execution agent. SOP:

1. Configure TWCC: module load cuda/12.3 nvhpc-24.11 gcc9; use g++, add -O3 -march=native; submit only via sbatch -A ACD115083 (never login node).
2. Classify each kernel’s roofline regime from its task type.

3. Implement concurrent CPU+GPU split (std::thread GPU + OpenMP CPU).
4. Calibrate $f=0$ and $f=1$; compute $f^* = R_{\text{gpu}}/(R_{\text{cpu}}+R_{\text{gpu}})$. Do NOT brute force.
5. Sweep f to validate the predicted optimum; gradient-correct on transfer.
6. Report mix vs pure-CPU vs pure-GPU.

Appendix D

Representative Agent Output

```

stencil (memory-bound): Rcpu=7.4 GB/s Rgpu=11.5 GB/s
predicted  $f^* = 0.609$  ; empirical best  $f = 0.600$ 
pure CPU 72.3ms | pure GPU 46.5ms | co-exec 43.9ms
speedup vs CPU 1.65x , vs GPU 1.06x      <-- TRUE MIX
WIN

gemm (compute-bound):  Rcpu=47 GFLOP/s Rgpu=2044
GFLOP/s
predicted  $f^* = 0.977$  ; empirical best  $f = 1.000$  (pure GPU)
mix collapses to GPU (43x faster than CPU) -- no split helps

```

Appendix E

Changes Made to the Agent

The agent diagnosed and fixed a hung run (missing -O3 made unoptimized MLP matmul exceed the wall-clock limit) and a shell bug, then added per-job timeout guards. It created the two micro-kernels (coexec_gemm.cu, coexec_stencil.cu), the ten full-benchmark co-execution drivers (coexec_3mm/mlp/oha/cnn/adi/ced/sbf/opf/spf/ppc.cu sharing coexec_common.h), the sweep harness (run_suite.sh) and analyzers (analyze_coexec.py, analyze_suite.py).

Reproducibility. All source code, the co-execution kernels, the analytical optimizer, the raw CSV sweeps (results/suite_sweep.csv, suite_results.json) and a step-by-step README.md are publicly available at <https://github.com/h3902340/padp-final>. The two-kernel core (Sec. VI) reproduces in under two minutes via `cd coexec && make && ./run_coexec.sh`; the full ten-benchmark sweep (Table I, Fig. 4) via `sbatch run_suite.sh` (about five minutes on one V100).